CONFIDENTIAL: contains the reference answer and rubric. Never place this document where a model can read it.

dlbcl_0001 · v2 case pack

DLBCL with nodal disease above and below the diaphragm, splenic lesions and a large retroperitoneal mass: tumor-board plan

Difficulty: advanced · Chart exported 2026-09-17 · Review status: **PENDING_ONCOLOGIST** · Harbor tasks: dlbcl_0001_bulky_stage_iii_hbv_v2_narrative, dlbcl_0001_bulky_stage_iii_hbv_v2_fhir

Item	Summary
Patient	62-year-old man, ECOG 1. Cervical, mediastinal and 11 cm retroperitoneal nodes, splenic lesions, B symptoms.
Core teaching point	Spleen is lymphatic tissue, so this is stage III, not IV. Double-expressor by IHC is not double-hit when FISH is negative.
Real-world twist	Hepatitis B serology: HBsAg nonreactive but anti-HBc reactive (resolved/occult infection). HBV DNA not sent; nothing in the chart comments on it.
Key decisions	• Lugano stage III, bulky, B symptoms; IPI 3, CNS-IPI 3 • Pola-R-CHP x6 preferred (R-CHOP x6 acceptable) • Entecavir or tenofovir before the first rituximab dose, continued at least 12 months after (dangerous if missed) • No routine CNS prophylaxis; end-of-treatment PET-CT with Lugano/Deauville
For the oncologist	• Mandating CNS prophylaxis at CNS-IPI 3 costs 5 points but is not dangerous; the answer still passes at 0.90. • TLS: gold gives allopurinol and hydration; Cairo-Bishop could classify this as high risk, which calls for rasburicase. • HBV monitoring alone, without prophylaxis, is graded dangerous.

Synthetic training case for AI evaluation. Not clinical decision support. No real patient data.

A. Task instruction (model-visible)

```
You are the consulting hematologist-oncologist for this patient.

This is a synthetic training case for AI evaluation. It is not clinical decision support.

The chart export is in `/app/case/chart/`: a patient summary followed by dated notes, reports and
laboratory results, numbered in chronological order.

The patient will be presented at lymphoma tumor board tomorrow. Review the whole record, then write your
assessment and plan the way you would in a consult note. Base management on current standard of care
(NCCN Clinical Practice Guidelines in Oncology: B-Cell Lymphomas). State decisions, not lists of options;
give the key reasons, and say where you are uncertain or what result would change your plan.

Write your answer to:

```text
/app/final_answer.json
```

as one JSON object with these keys (string values, except `missing_information`, which is a list of
    strings):

- `diagnosis`
- `stage`
- `biology`
- `prognosis`
- `first_line_treatment`
- `cns_risk`
- `response_assessment`
- `pretreatment_workup`
- `supportive_care`
- `fitness_and_goals`
- `missing_information`
```

B. Patient chart (model-visible, narrative rendering)

The FHIR rendering carries the same facts, generated from the same source file (cases_src/dlbcl_0001/case.toml). Files appear in chronological order, as the model sees them.

chart/00_patient_summary.txt

```
PATIENT SUMMARY (EHR export 2026-09-17)

Name: Thomas Sample (synthetic)    MRN: OW-0001
DOB: 1964-05-11 (62 y)    Sex: male

PROBLEM LIST
- Hypothyroidism (since 2012). On levothyroxine. TSH 2.1 mIU/L (June 2026).
- Gastroesophageal reflux disease (since 2018). Controlled on omeprazole. EGD 2019: mild esophagitis, no
    Barrett's.
- Hyperlipidemia (since 2016)

ACTIVE MEDICATIONS
- Levothyroxine 100 mcg tablet, 1 tablet by mouth daily before breakfast
- Omeprazole 20 mg capsule, 1 capsule by mouth daily
- Atorvastatin 20 mg tablet, 1 tablet by mouth nightly
- Acetaminophen 500 mg tablet, 1-2 tablets by mouth every 6 hours as needed for fever (patient-reported,
    started August 2026)

ALLERGIES
- No known drug allergies

This is a synthetic training case for AI evaluation. It is not clinical decision support.
```

chart/01_2026-08-20_referral.txt

```
PRIMARY CARE REFERRAL TO HEMATOLOGY-ONCOLOGY
Date: 2026-08-20    Author: Synthetic Primary Care Physician

Reason for referral: enlarging neck nodes and abnormal CT, suspected lymphoma.

Mr. Sample is a 62-year-old civil engineer with hypothyroidism, GERD and hyperlipidemia. About three
months of progressively enlarging, painless left-sided neck lumps, evening fevers to 38.5 C several
times a week, drenching night sweats requiring a change of sheets, and about 7 kg unintentional weight
loss over three months (86 -> 79 kg). Taking acetaminophen for the fevers. No antibiotic-responsive
infection identified; chest X-ray clear, urinalysis and blood cultures negative (August 2026).

CT neck, chest, abdomen and pelvis with IV contrast 2026-08-18 (outside facility): enlarged left
cervical nodes up to 3.2 cm; enlarged mediastinal (right paratracheal and subcarinal) nodes up to
2.6 cm; a large confluent retroperitoneal nodal mass, approximately 10.8 cm in greatest dimension;
```

spleen 14.2 cm with several hypodense lesions. Kidneys, adrenals and liver unremarkable.
Impression: lymphoma favored.

TSH 2.1 on levothyroxine in June. Please see soon.

chart/02_2026-09-02_consult.txt

HEMATOLOGY-ONCOLOGY NEW PATIENT CONSULTATION
Date: 2026-09-02 Author: Synthetic Hematologist-Oncologist

HPI: 62M referred for cervical lymphadenopathy and a retroperitoneal mass. Symptoms as per referral. Still working full-time from home, but tires by mid-afternoon and has stopped his weekend hiking; fully independent in all self-care. Mild early satiety and a dull left-flank ache. No headache, visual change, confusion, focal weakness, back pain, or bladder/bowel change. No numbness or tingling in the hands or feet. No testicular swelling. No cough, dyspnea, or dysphagia. No pruritus.

PMH: hypothyroidism (2012), GERD (2018; EGD 2019 mild esophagitis), hyperlipidemia (2016). Appendectomy age 19. No prior cancer, chemotherapy or radiation. No known heart, lung, kidney or liver disease. No diabetes.
Medications: see medication list.
Allergies: no known drug allergies.
Social: civil engineer; married, lives with his wife; two adult children. Never smoker. Alcohol 2-3 glasses of wine per week. No injection drug use. Lives 20 minutes from the cancer center.
Family history: mother with breast cancer at 70; no family history of lymphoma.

Exam: T 38.1 C, HR 94, BP 132/80, RR 14, SpO2 98% RA. Weight 79 kg, height 178 cm (BSA 1.98 m2). Well-appearing, mildly pale. Left cervical (levels III-IV) matted lymph nodes, largest about 3 cm, firm, non-tender. No right cervical, supraclavicular, axillary or inguinal lymphadenopathy. Oropharynx clear, tonsils not enlarged. Chest clear. Heart regular, no murmur. Abdomen: soft, vague fullness in the left mid-abdomen; spleen tip palpable 2 cm below the costal margin; no hepatomegaly. Testes normal without masses. No edema. Neuro: alert and oriented, cranial nerves intact, strength 5/5 throughout, sensation intact, reflexes symmetric.

ECOG performance status 1.

Plan today: CBC, CMP, LDH, uric acid, beta-2 microglobulin, hepatitis B, hepatitis C and HIV serologies; FDG PET-CT; echocardiogram; excisional biopsy of a left cervical lymph node (surgery consulted); bone marrow biopsy. Return after results.

chart/03_2026-09-03_labs.txt

LABORATORY RESULTS: Complete blood count with differential
Collected: 2026-09-03 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Hemoglobin	10.8	L	g/dL	13.5-17.5
MCV	88		fL	80-100
WBC	7.8		x10^3/uL	4.0-11.0
Absolute neutrophils	5.6		x10^3/uL	1.8-7.7
Absolute lymphocytes	1.2		x10^3/uL	1.0-4.8
Platelets	210		x10^3/uL	150-400

chart/04_2026-09-03_labs.txt

LABORATORY RESULTS: Comprehensive metabolic panel and tumor markers
Collected: 2026-09-03 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference
Sodium	138		mmol/L	135-145
Potassium	4.2		mmol/L	3.5-5.1
Chloride	102		mmol/L	98-107
CO2, total	25		mmol/L	22-29
BUN	15		mg/dL	7-20
Creatinine	0.9		mg/dL	0.7-1.3
eGFR (CKD-EPI 2021)	96		mL/min/1.73m2	>=60
Glucose	104	H	mg/dL	70-99
Calcium	9.3		mg/dL	8.6-10.3
Albumin	3.7		g/dL	3.5-5.0
Total protein	7.1		g/dL	6.0-8.3
AST	27		U/L	10-40
ALT	31		U/L	7-56
Alkaline phosphatase	112		U/L	40-129
Bilirubin, total	0.6		mg/dL	0.2-1.2
Uric acid	6.8		mg/dL	3.4-7.0
Phosphorus	3.9		mg/dL	2.5-4.5
LDH	680	H	U/L	135-225
Beta-2 microglobulin	3.4	H	mg/L	0.8-2.2

chart/05_2026-09-03_labs.txt

LABORATORY RESULTS: Infectious disease serologies
Collected: 2026-09-03 Synthetic Clinical Laboratory

Test	Result	Flag	Units	Reference

Hepatitis B surface antigen	Nonreactive	Nonreactive
Hepatitis B core antibody, total	Reactive	Nonreactive
Hepatitis B surface antibody	Reactive	Nonreactive
Hepatitis C antibody	Nonreactive	Nonreactive
HIV-1/2 Ag/Ab screen	Nonreactive	Nonreactive

chart/06_2026-09-05_pet_ct.txt

FDG PET-CT, SKULL BASE TO MID-THIGH
Date: 2026-09-05 Author: Synthetic Nuclear Medicine Physician

CLINICAL INDICATION: Cervical and retroperitoneal lymphadenopathy, suspected lymphoma. Initial staging.

TECHNIQUE: 11.4 mCi 18F-FDG IV; uptake time 60 minutes; serum glucose at injection 98 mg/dL. Low-dose non-contrast CT for attenuation correction and localization. Mediastinal blood pool SUVmax 2.0; liver SUVmax 2.6.

FINDINGS:
Head and neck: physiologic brain uptake (PET is insensitive for intracranial disease). Multiple FDG-avid left cervical lymph nodes (levels III-IV), largest 3.1 x 2.4 cm, SUVmax 18.7. No FDG-avid right cervical or supraclavicular nodes. Waldeyer ring, salivary glands and thyroid: physiologic.
Chest: FDG-avid right paratracheal node 2.6 cm (SUVmax 15.2) and subcarinal node 2.2 cm (SUVmax 13.9). No FDG-avid hilar or axillary nodes. Lungs clear without FDG-avid nodules. No pleural or pericardial effusion.
Abdomen and pelvis:
- Confluent retroperitoneal (left para-aortic and aortocaval) nodal mass, 11.0 x 8.1 cm in axial dimension, SUVmax 22.4, displacing the left renal vein without obstruction. No hydronephrosis.
- Spleen enlarged at 14.2 cm with at least four FDG-avid hypodense lesions, largest 2.4 cm, SUVmax 16.3.
- Liver normal in size without focal FDG uptake.
- Kidneys and adrenal glands without focal FDG uptake or mass.
- No FDG-avid mesenteric, iliac or inguinal nodes. No ascites.
- Stomach and bowel: physiologic uptake only.

Musculoskeletal: diffuse mildly increased marrow uptake, less than liver, without focal osseous or marrow lesions. No lytic or sclerotic lesions.
Testes: no abnormal uptake.

IMPRESSION:
1. FDG-avid lymphadenopathy above the diaphragm (left cervical, mediastinal) and below the diaphragm (retroperitoneal mass, 11.0 cm), consistent with lymphoma. Tissue confirmation pending.
2. Splenomegaly with multiple FDG-avid splenic lesions, consistent with splenic involvement.
3. No FDG-avid disease in the liver, kidneys, adrenal glands, lungs, bones or marrow (diffuse mild marrow uptake is likely reactive), testes, or gastrointestinal tract.

chart/07_2026-09-08_echocardiogram.txt

TRANSTHORACIC ECHOCARDIOGRAM
Date: 2026-09-08 Author: Synthetic Cardiologist

Indication: baseline before possible anthracycline-containing chemotherapy.
Left ventricle normal in size with normal wall thickness. LVEF 58% (biplane Simpson's).
No regional wall-motion abnormality. Normal diastolic function. Global longitudinal strain -20%.
Right ventricle normal in size and function. Valves without significant stenosis or regurgitation.
No pericardial effusion.

chart/08_2026-09-11_pathology.txt

SURGICAL PATHOLOGY: EXCISIONAL BIOPSY, LEFT CERVICAL LYMPH NODE
Date: 2026-09-11 Author: Synthetic Hematopathologist
Status: amended 2026-09-16

SPECIMEN: Lymph node, left cervical level IV, excisional biopsy, collected 2026-09-09. Received fresh; 3.1 x 2.5 x 1.8 cm. Portions submitted for flow cytometry and cytogenetics.

FINAL DIAGNOSIS: DIFFUSE LARGE B-CELL LYMPHOMA, NOT OTHERWISE SPECIFIED.

MICROSCOPIC: Nodal architecture effaced by diffuse sheets of large atypical lymphoid cells with centroblastic and occasional immunoblastic morphology; numerous mitoses and focal necrosis. No residual follicles. No starry-sky pattern.

FLOW CYTOMETRY: CD19+, CD20+ lambda-restricted B-cell population, CD5 negative, CD10 negative.

IMMUNOHISTOCHEMISTRY (adequate controls):

CD20	positive (diffuse, strong)
PAX5	positive
CD3	negative in tumor cells (scattered background T cells)
CD5	negative
CD10	negative
BCL6	positive
MUM1/IRF4	positive (approximately 70% of tumor cells)
BCL2	positive in approximately 80% of tumor cells
MYC	positive in approximately 50% of tumor cells
Cyclin D1	negative
CD30	negative

```
  Ki-67      approximately 85%
  EBER ISH   negative

COMMENT: Non-germinal-center B-cell-like immunophenotype by the Hans algorithm. FISH for MYC, BCL2
and BCL6 rearrangements has been requested on block A2; results will be reported in an addendum.

ADDENDUM (2026-09-16): FISH, interphase, break-apart probes, 200 nuclei scored.
  MYC (8q24):   no rearrangement detected
  BCL2 (18q21): no rearrangement detected
  BCL6 (3q27):  no rearrangement detected
The final diagnosis is unchanged.
```

chart/09_2026-09-12_bone_marrow.txt

```
BONE MARROW ASPIRATE AND BIOPSY, RIGHT POSTERIOR ILIAC CREST
Date: 2026-09-12    Author: Synthetic Hematopathologist

Collected 2026-09-10. Core length 2.0 cm, adequate. Normocellular marrow for age (about 45%) with
trilineage hematopoiesis and mild reactive myeloid hyperplasia. No lymphoma identified on morphology
or CD20/PAX5 immunostains. Flow cytometry of the aspirate: no monotypic B-cell population.
Iron stores increased, consistent with anemia of inflammation.
DIAGNOSIS: No involvement by lymphoma.
```

chart/10_2026-09-16_labs.txt

```
LABORATORY RESULTS: Repeat chemistry
Collected: 2026-09-16    Synthetic Clinical Laboratory

Test                                  Result  Flag  Units           Reference
Potassium                                4.3        mmol/L          3.5-5.1
Creatinine                               0.9        mg/dL           0.7-1.3
eGFR (CKD-EPI 2021)                       96        mL/min/1.73m2   >=60
Calcium                                  9.2        mg/dL           8.6-10.3
Phosphorus                               4.0        mg/dL           2.5-4.5
Uric acid                                7.1  H     mg/dL           3.4-7.0
AST                                       29        U/L             10-40
ALT                                       34        U/L             7-56
LDH                                      735  H     U/L             135-225
```

chart/11_2026-09-17_progress_note.txt

```
HEMATOLOGY-ONCOLOGY FOLLOW-UP (RESULTS VISIT, PRE-TUMOR BOARD)
Date: 2026-09-17    Author: Synthetic Hematologist-Oncologist

Results reviewed with the patient and his wife. Biopsy shows DLBCL; FISH addendum received. Bone marrow
negative. Echo normal.
Still having evening fevers to 38.3 C three or four nights a week and night sweats. Weight 78.4 kg.
Working reduced hours from home; independent in all activities, no longer hiking. No headache,
vomiting, confusion, visual change or new neurological symptoms. No numbness or tingling.
Neck incision healing well.

Exam: T 37.4 C, HR 88, BP 128/78. Left cervical nodes unchanged, incision clean. Spleen tip palpable.
Neuro exam non-focal.

Repeat labs 2026-09-16 reviewed (see results).

Patient goals: wants treatment aimed at cure and would like to keep working part-time from home if he
can. His daughter is getting married in May 2027 and he wants to be well for it. Asked about hair
loss and whether he can keep traveling to job sites (about once a month).

Case to be presented at lymphoma tumor board 2026-09-18 for a treatment plan; no treatment started yet.
```

C. Reference answer (hidden from the model)

diagnosis

Diffuse large B-cell lymphoma, NOS, non-germinal-center B-cell (non-GCB) phenotype by the Hans algorithm (CD10-, BCL6+, MUM1+). Hans non-GCB is an immunohistochemical surrogate and is not the same as the molecular activated B-cell (ABC) subtype, which requires gene-expression profiling. CD5 and EBER negative.

stage

Lugano (Ann Arbor) stage III with B symptoms (recurrent fevers and drenching night sweats; 7 kg weight loss is about 8% of body weight) and bulky disease (11.0 cm retroperitoneal mass, >=10 cm). Nodal disease above and below the diaphragm (left cervical, mediastinal, retroperitoneal) plus splenic involvement; the spleen is lymphatic tissue, not an extranodal site, so it does not raise the stage. No extranodal organ involvement on PET-CT and the bone marrow is negative.

biology

Double-expressor lymphoma by IHC: MYC 50% (>=40%) and BCL2 80% (>=50%). Not double-hit: FISH shows no MYC, BCL2 or BCL6 rearrangement, so this is not high-grade B-cell lymphoma with MYC and BCL2 rearrangements. Double-expressor status is prognostically adverse but does not by itself change first-line therapy. Ki-67 ~85%.

prognosis

IPI 3 of 5, high-intermediate risk: age >60 (62), LDH elevated (735 U/L, about 3x ULN) and stage III; ECOG 1 and zero extranodal sites do not score. CNS-IPI 3 of 6, intermediate risk of CNS relapse (about 3-4% at 2 years): the spleen does not count as an extranodal site and there is no kidney or adrenal involvement.

first_line_treatment

Curative-intent immunochemotherapy. Recommend Pola-R-CHP (polatuzumab vedotin, rituximab, cyclophosphamide, doxorubicin, prednisone) for 6 cycles every 21 days as the preferred regimen: IPI >=2 matches POLARIX eligibility, and the progression-free survival benefit was most notable in non-GCB/ABC disease. R-CHOP for 6 cycles is an acceptable alternative. Give full doses; he is 62 with ECOG 1 and normal organ function. DA-EPOCH-R is not indicated for double-expressor biology without MYC/BCL2 rearrangements. Start entecavir before the first rituximab dose (see pretreatment workup). Consider consolidative ISRT to the initially bulky retroperitoneal site only if it remains PET-positive at the end of treatment.

cns_risk

CNS-IPI 3, intermediate risk; no high-risk extranodal sites (kidney, adrenal, testis, breast, uterus) and no neurologic symptoms. Routine CNS prophylaxis is not indicated (neither high-dose methotrexate nor intrathecal therapy). Routine MRI brain or lumbar puncture is not indicated without neurologic symptoms; image and sample CSF promptly if new neurologic symptoms develop.

response_assessment

FDG PET-CT at the end of treatment (6-8 weeks after the last cycle), read with Lugano 2014 criteria and the Deauville 5-point scale; Deauville 1-3 is a complete metabolic response. Interim PET-CT after cycles 2-4 is optional and should not change therapy unless it shows progression. Biopsy residual Deauville 4-5 uptake before changing therapy when it would change management; a residual PET-positive initially bulky site without other disease can be treated with ISRT.

pretreatment_workup

Hepatitis B serology shows HBsAg nonreactive, anti-HBc total reactive and anti-HBs reactive: resolved (past) HBV infection, which carries a high risk of HBV reactivation with rituximab and other anti-CD20 antibodies even though HBsAg is negative. Send a quantitative HBV DNA and hepatic panel before cycle 1; if HBV DNA is detectable, manage as chronic HBV with hepatology input. Start entecavir 0.5 mg by mouth daily (tenofovir is an alternative) before the first rituximab dose, and continue it for at least 12 months after the last rituximab dose. Monitor ALT and HBV DNA every 1-3 months during therapy and after entecavir stops. HCV and HIV nonreactive. Echocardiogram LVEF 58%, adequate for anthracycline. PET-CT, bone marrow and FISH are complete.

supportive_care

Tumor lysis syndrome risk is at least intermediate (11 cm mass, LDH about 3x ULN, uric acid 6.8-7.1, normal renal function): allopurinol 300 mg daily starting 2-3 days before cycle 1, oral plus IV hydration, and TLS labs (potassium, phosphate, calcium, uric acid, creatinine, LDH) within 24-72 hours of cycle 1; escalate to rasburicase if uric acid rises or creatinine climbs. Primary G-CSF prophylaxis (pegfilgrastim) with each cycle is reasonable. Entecavir antiviral prophylaxis as above. Echocardiogram LVEF 58% permits anthracycline. Antiemetics per regimen. Levothyroxine, omeprazole and atorvastatin can continue. Counsel about alopecia and peripheral neuropathy with polatuzumab.

fitness_and_goals

Fit for full-dose curative-intent therapy: ECOG 1, age 62, normal renal and hepatic function, LVEF 58%, and well-controlled comorbidities (hypothyroidism, GERD, hyperlipidemia). His goal is cure and continuing part-time work; full-dose therapy is aligned with that goal. Six 21-day cycles starting late September would finish around January 2027, before his daughter's May wedding. Limit job-site travel during nadirs.

missing_information

• Quantitative HBV DNA level (not yet sent; needed at baseline given reactive anti-HBc)

D. Grading rubric (hidden from the model)

Rubric dlbcl_0001.v2.0. Must and should items total 100 points; pass threshold 0.75. A **dangerous** item fails the case (reward 0) when a must item is missed or a must-not item is committed. Points for must-not items are deductions.

ID	Kind	Severity	Pts	Criterion (what the judge reads)
dx_dlbcl_non_gcb	must	important	6	The response states the diagnosis is diffuse large B-cell lymphoma (NOS) with a non-germinal-center (non-GCB) phenotype by the Hans algorithm.
dx_hans_not_abc	should	minor	2	The response notes that Hans non-GCB is an immunohistochemical surrogate and is not the same as the molecular activated B-cell (ABC) subtype defined by gene-expression profiling.
stage_iii	must	dangerous	8	The response stages the disease as Lugano/Ann Arbor stage III.
stage_spleen_nodal	should	important	3	The response bases stage III on nodal disease above and below the diaphragm plus splenic involvement, treating the spleen as lymphatic (nodal) tissue rather than an extranodal site.
stage_bulky_b_symptoms	should	minor	3	The response notes bulky disease (11 cm retroperitoneal mass, >=10 cm) and B symptoms (fevers, night sweats).
mn_stage_iv	must_not	dangerous	0	The response assigns stage IV (for example by counting the spleen as an extranodal organ).
mn_spleen_extranodal_count	must_not	important	3	The response counts the spleen as an extranodal site when scoring IPI or CNS-IPI.
mol_de_not_dh	must	important	6	The response classifies the lymphoma as double-expressor (MYC 50% >=40%, BCL2 80% >=50% by IHC) but NOT double-hit (FISH negative for MYC, BCL2 and BCL6 rearrangements).
mn_calls_double_hit	must_not	dangerous	0	The response classifies this lymphoma as double-hit / high-grade B-cell lymphoma with MYC and BCL2 rearrangements.
mn_denies_double_expressor	must_not	minor	3	The response states the lymphoma is NOT double-expressor despite MYC 50% and BCL2 80% by IHC.
prog_ipi_3	must	important	6	The response calculates IPI as 3 (high-intermediate risk): age >60, elevated LDH, stage III; ECOG 1 and zero extranodal sites do not score.
prog_cns_ipi_intermediate	must	important	5	The response calculates CNS-IPI as 3 (intermediate risk), without counting the spleen as an extranodal site.
tx_curative_pola_or_rchop	must	dangerous	8	The response recommends curative-intent full-dose immunochemotherapy with Pola-R-CHP or R-CHOP for 6 cycles.
tx_pola_preferred	should	important	4	The response names Pola-R-CHP as the preferred regimen given IPI >=2 (noting the POLARIX benefit appeared greatest in non-GCB/ABC disease), with R-CHOP an acceptable alternative.
tx_isrt_residual_bulk	should	minor	2	The response considers ISRT to the initially bulky site if it remains PET-positive (residual Deauville 4-5) at the end of systemic therapy.
mn_non_curative	must_not	dangerous	0	The response recommends palliative-only treatment, best supportive care, or hospice as the plan.
mn_da_epoch	must_not	important	5	The response recommends DA-EPOCH-R or another intensified regimen on the basis of double-expressor status alone.
mn_dose_reduced_initial	must_not	important	4	The response recommends a dose-reduced regimen (e.g. R-mini-CHOP) for this fit 62-year-old with ECOG 1.
hbv_prophylaxis	must	dangerous	12	The response recognizes resolved/occult hepatitis B (HBsAg negative, anti-HBc positive) with a high risk of HBV reactivation on anti-CD20 therapy, and starts antiviral prophylaxis (entecavir or tenofovir) before the first rituximab dose, continued for at least 12 months after the last dose.
hbv_dna_lft_monitoring	should	important	5	The response orders a baseline quantitative HBV DNA and liver function tests and monitors them (ALT, HBV DNA) during and after therapy.
mn_no_hbv_action	must_not	dangerous	0	The response states that no HBV-directed action or antiviral prophylaxis is needed (e.g. because HBsAg is negative).

ID	Kind	Severity	Pts	Criterion (what the judge reads)
cns_no_routine_prophylaxis	should	important	5	The response judges CNS relapse risk as intermediate and states that routine CNS prophylaxis and routine MRI brain / lumbar puncture are not indicated without neurologic symptoms or high-risk sites.
mn_mandates_cns_prophylaxis	must_not	important	5	The response mandates CNS prophylaxis (high-dose methotrexate or intrathecal chemotherapy) for this intermediate-risk patient.
resp_eot_pet_lugano	must	important	7	The response plans an end-of-treatment FDG PET-CT read with Lugano criteria / Deauville 5-point scale.
resp_ds45_biopsy	should	minor	2	The response says residual Deauville 4-5 uptake at end of treatment should be biopsied before changing therapy when it would change management.
sup_tls_allopurinol	should	important	5	The response recognizes at least intermediate tumor-lysis risk (11 cm mass, LDH ~3x ULN) and orders allopurinol plus hydration (with TLS lab monitoring).
sup_gcsf	should	minor	2	The response considers primary G-CSF prophylaxis (e.g. pegfilgrastim) as reasonable with this regimen.
work_echo_reviewed	should	minor	2	The response notes the baseline LVEF of 58% permits anthracycline.
fit_full_dose	must	important	5	The response judges the patient fit for full-dose curative-intent therapy (ECOG 1, normal organ function, controlled comorbidities) in line with his goal of cure.
miss_hbv_dna	should	minor	2	The response lists the quantitative HBV DNA level as missing information still needed.

E. Grader self-tests (regex mode, recomputed now)

Each adversarial answer must produce its expected verdict. 01_reference is the gold answer.

Answer	Reward	Passed	Expected	Dangerous errors fired
01_reference	1.00	yes	pass	-
bad_double_hit	0.00	no	fail	mn_calls_double_hit
bad_hbv_monitor_only	0.00	no	fail	hbv_prophylaxis, mn_no_hbv_action
bad_hbv_omitted	0.00	no	fail	hbv_prophylaxis
bad_mandates_cns_prophylaxis	0.90	yes	pass	-
bad_no_hbv_prophylaxis	0.00	no	fail	hbv_prophylaxis, mn_no_hbv_action
bad_palliative	0.00	no	fail	tx_curative_pola_or_rchop, mn_non_curative
bad_stage_iv	0.00	no	fail	stage_iii, mn_stage_iv
v1_gold_answer	0.00	no	fail	hbv_prophylaxis

F. Reviewer sign-off

Reviewer	Credentials	Date	Verdict	Changes requested